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Red Hat Quay

Enterprise Container Registry

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Agenda

- What is Red Hat Quay
- Enterprise capabilities
- Architecture at a glance
- Security: access, scanning, and compliance
- Replication and high availability
- OpenShift integration
- Operations and best practices
- Summary

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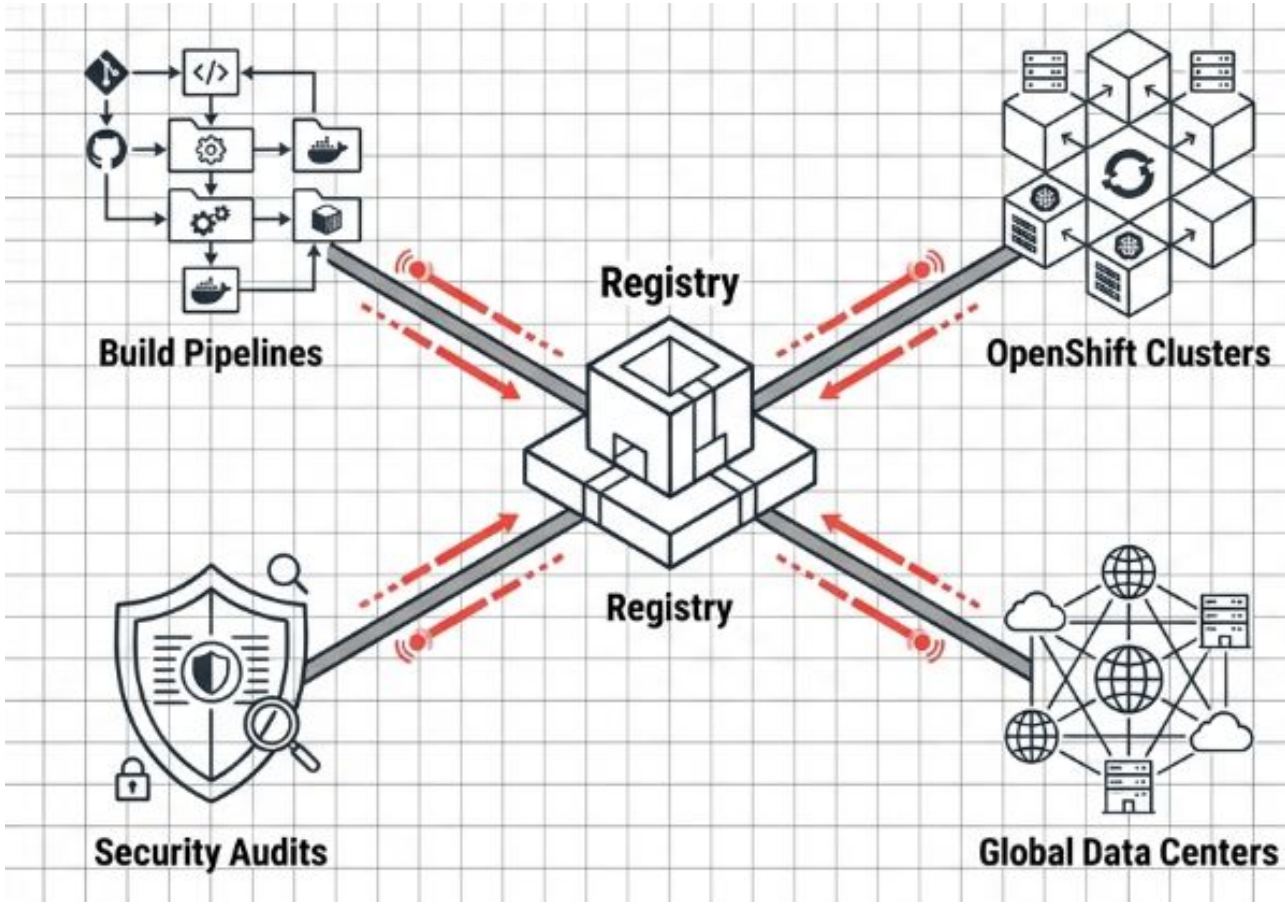
Introduction



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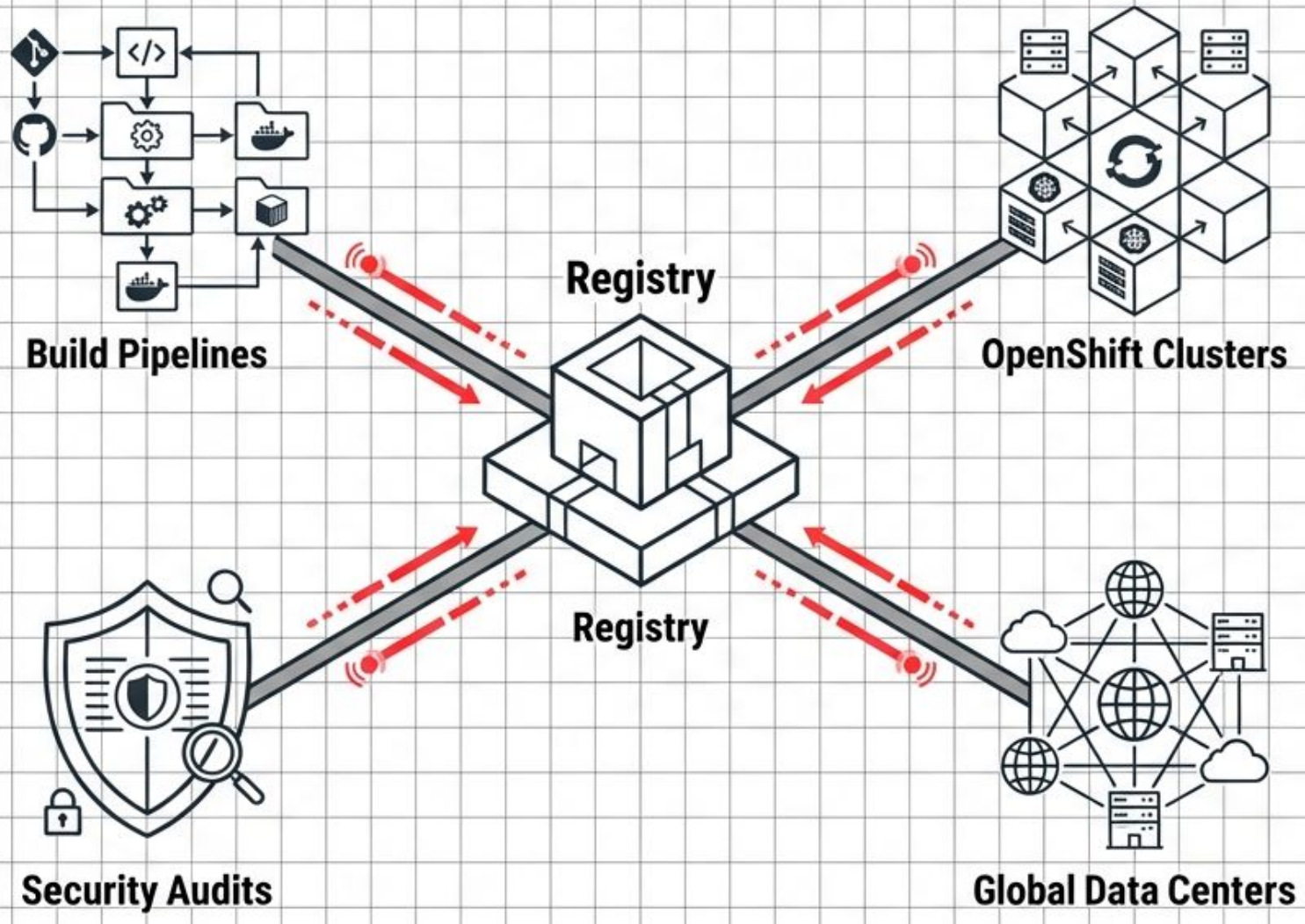
What Is Red Hat Quay

- Private registry for OCI images, Helm charts, and related artifacts
- Web UI, API, and CLI (podman/skopeo integration)
- Role-based access control for organizations, teams, and repositories
- Vulnerability scanning and policy hooks for secure supply chains
- Supported by Red Hat for production OpenShift and hybrid deployments

The registry is the active nexus of the Kubernetes architecture



A modern platform is only as resilient as the registry that feeds it.

- Availability: Global, low-latency access across oceans.
- Automation: Code changes trigger automated rebuilds.
- Security: Relentless vulnerability scanning.
- Resilience: Operates flawlessly from public clouds to subterranean airgapped data centers.

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Why Use an Enterprise Registry

- Central governance of images across teams and clusters
- Air-gapped and disconnected mirror workflows
- Audit trails: who pushed, pulled, or changed policy
- Integration with CI/CD and GitOps promotion pipelines
- Consistent experience from dev through production

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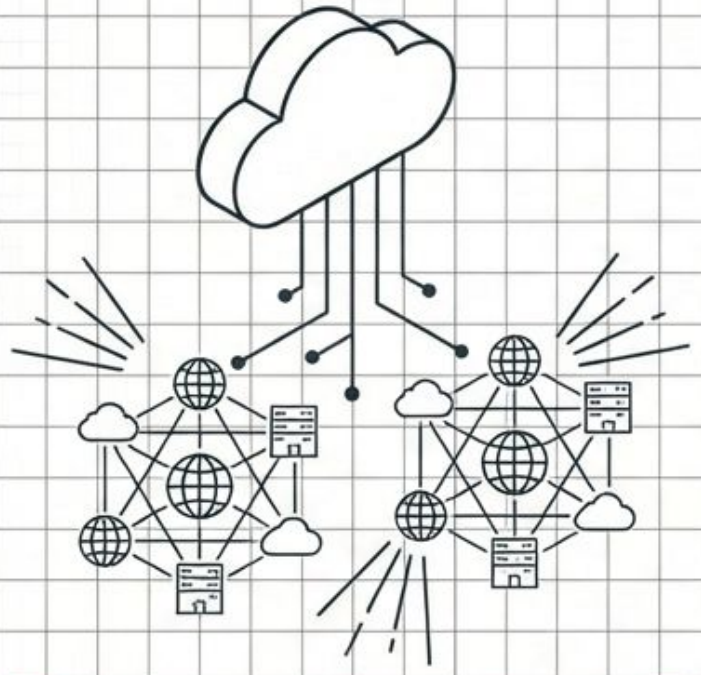
Enterprise Capabilities

Key Features

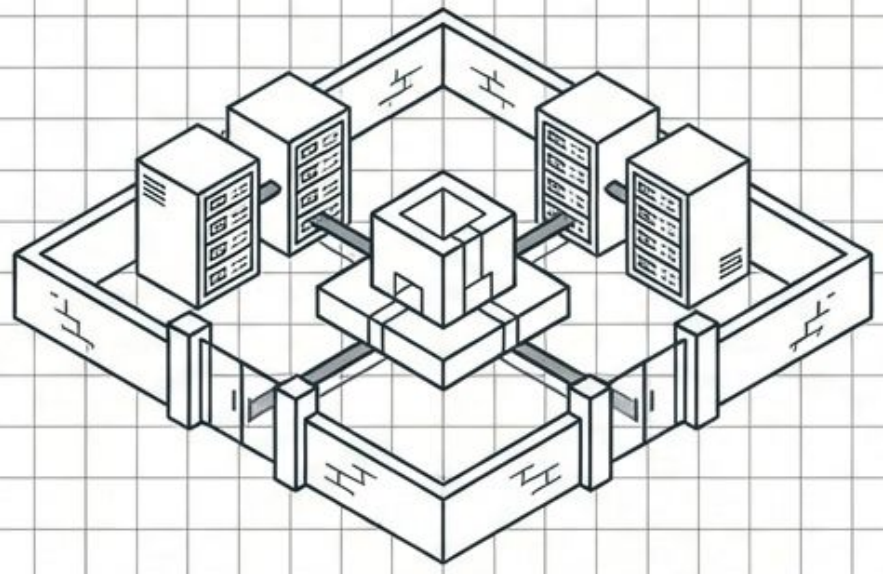
- Organizations and teams with fine-grained repository permissions
- Image mirroring and robust garbage collection
- Optional Clair-based vulnerability scanning (deployment dependent)
- Geo-replication for read-optimized multi-site distribution
- Quay Bridge Operator for OpenShift-native installation and lifecycle

Three distinct deployment topologies share a single registry DNA

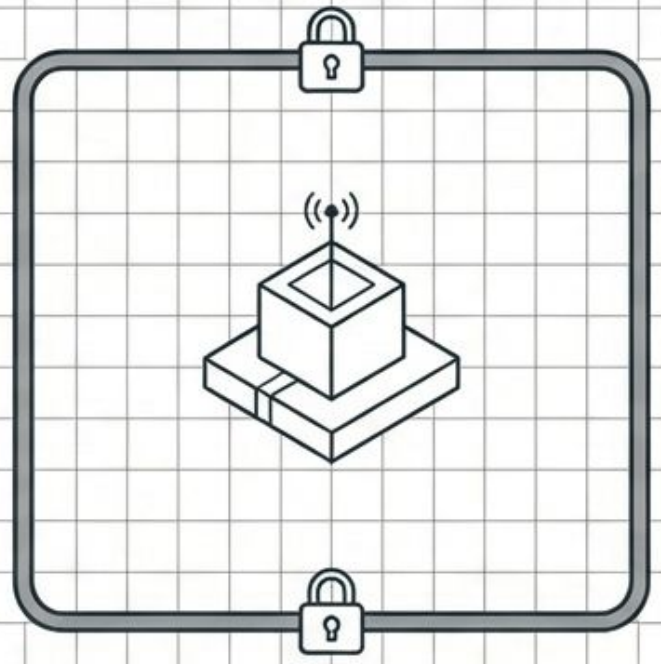
Quay.io



Red Hat Quay



Mirror Registry for OpenShift



Unified Ecosystem

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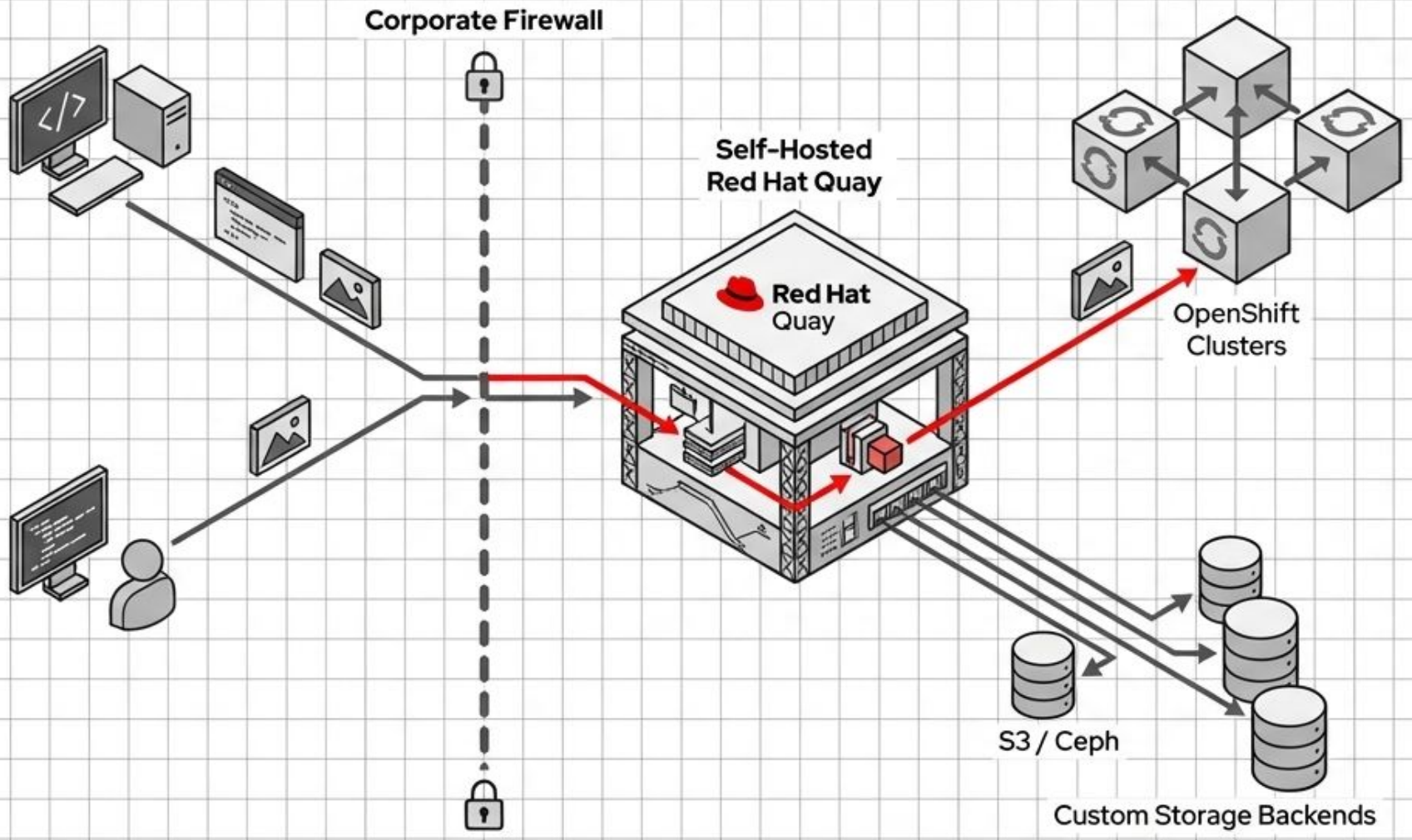
Red Hat Quay delivers absolute data sovereignty for the enterprise



Self-hosted for total data sovereignty.

Highly available and massively scalable.

Deep integration with OpenShift and enterprise identity providers.

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Architecture

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Architecture Overview

- Core API and registry service (pull/push protocol)
- Metadata store (database) for tags, manifests, and permissions
- Object storage backend: S3-compatible, Azure Blob, GCS, or local (dev)
- Background workers: build triggers, mirroring, garbage collection
- Optional Redis for caching and task queues in HA setups

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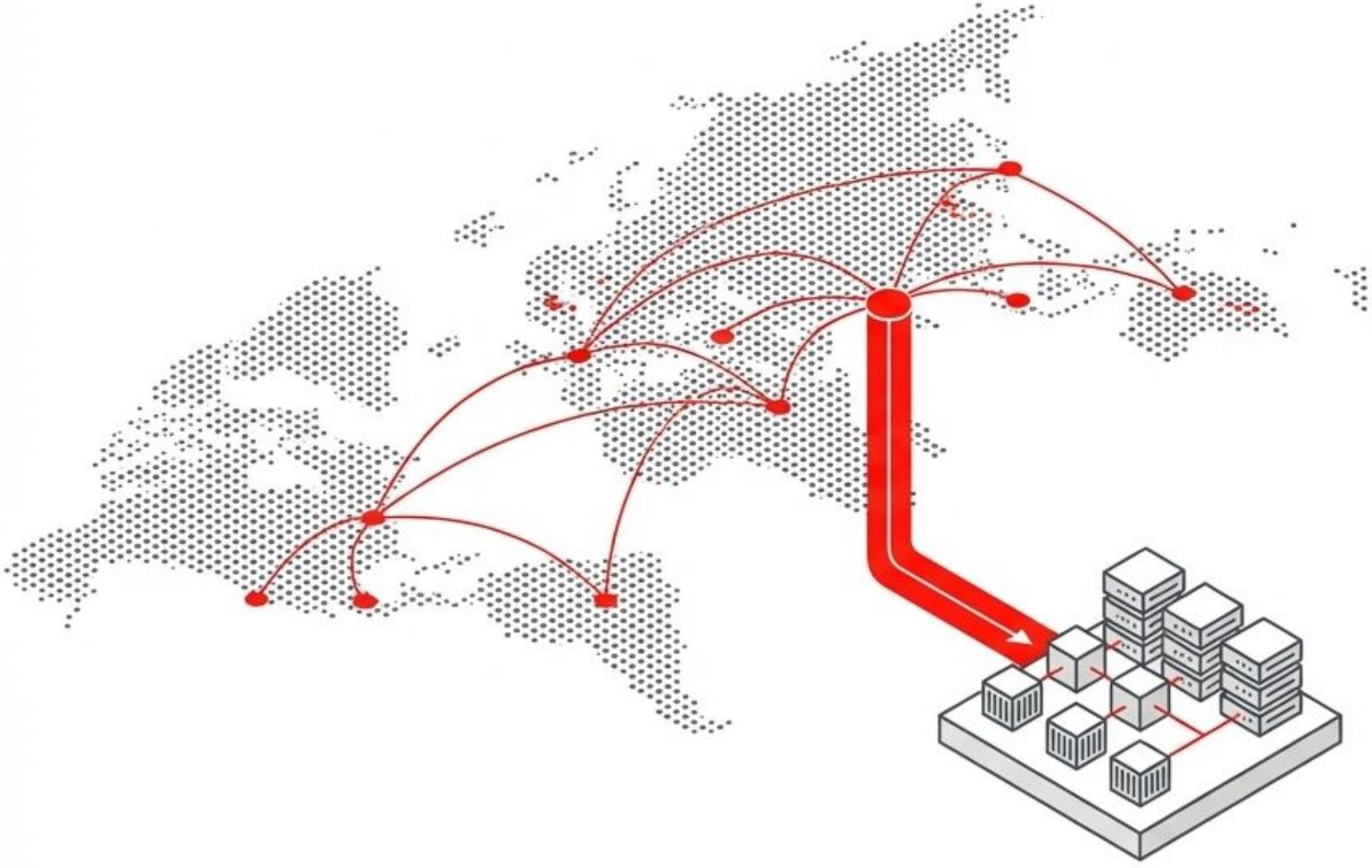
Quay.io provides planet-scale delivery with zero operational overhead



Hosted and **managed** entirely by Red Hat.

Stores millions of public images and tags.

The primary upstream source for OpenShift release images and Operators.

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Security

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Security and Compliance

- OAuth, LDAP, and OIDC integration for enterprise identity
- Robot accounts and tokens for automation with least privilege
- Vulnerability reports per image and policy gates in pipeline
- Content trust and signing workflows (cosign integration patterns)
- TLS everywhere; secrets managed via Kubernetes or vault patterns

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Replication and HA

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Replication and Availability

- Geo-replication: distribute images to regional Quay instances
- Mirroring rules sync selected repositories on a schedule
- High availability: multiple Quay pods, load balancer, resilient DB
- Backup and restore: database and storage are part of DR planning
- Test failover and restore procedures before you need them

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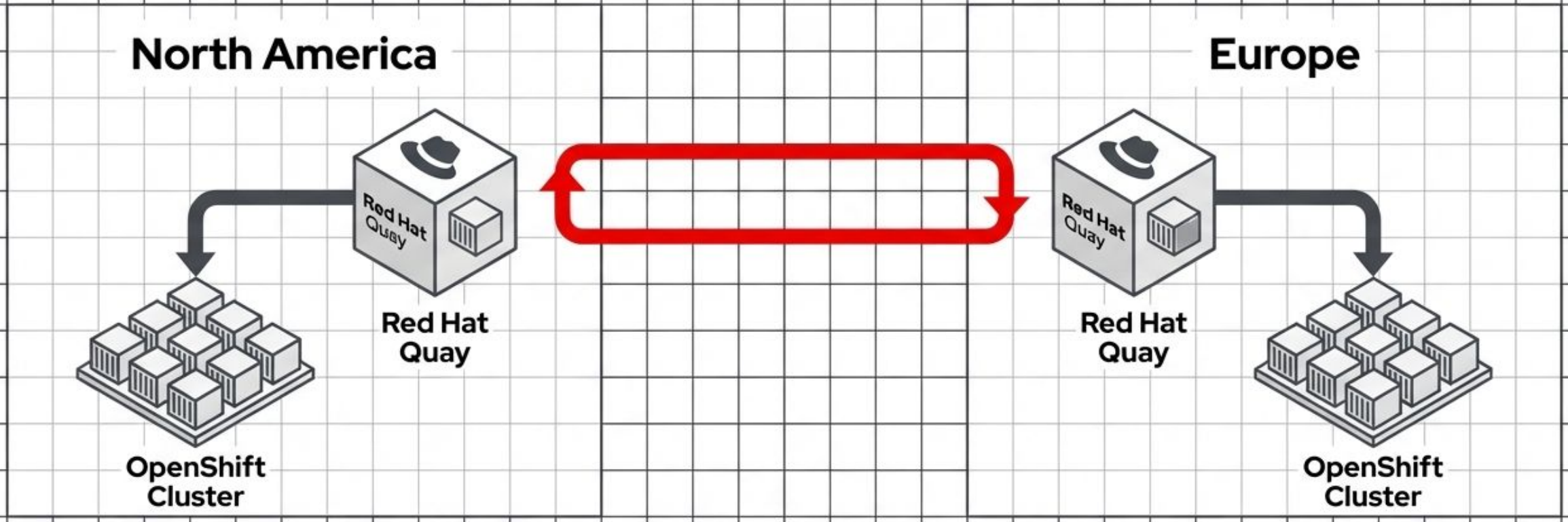
Geo-replication ensures local performance with global consistency



A single, globally consistent namespace synced automatically.

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OpenShift Integration

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Quay on OpenShift

- Quay Operator deploys and upgrades Quay on OpenShift
- Image streams and pull secrets wired to cluster projects
- OpenShift Pipelines can push and promote through Quay stages
- Centralized registry URL for many clusters in an estate
- Align RBAC with OpenShift projects and team structure

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Operations

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Best Practices

- Size storage for growth; monitor blob usage and GC windows
- Separate dev/stage/prod orgs or repositories with promotion rules
- Rotate robot tokens and integrate with secret management
- Monitor Quay health, DB performance, and scanning pipeline backlog
- Document restore runbooks and test upgrades in lower environments

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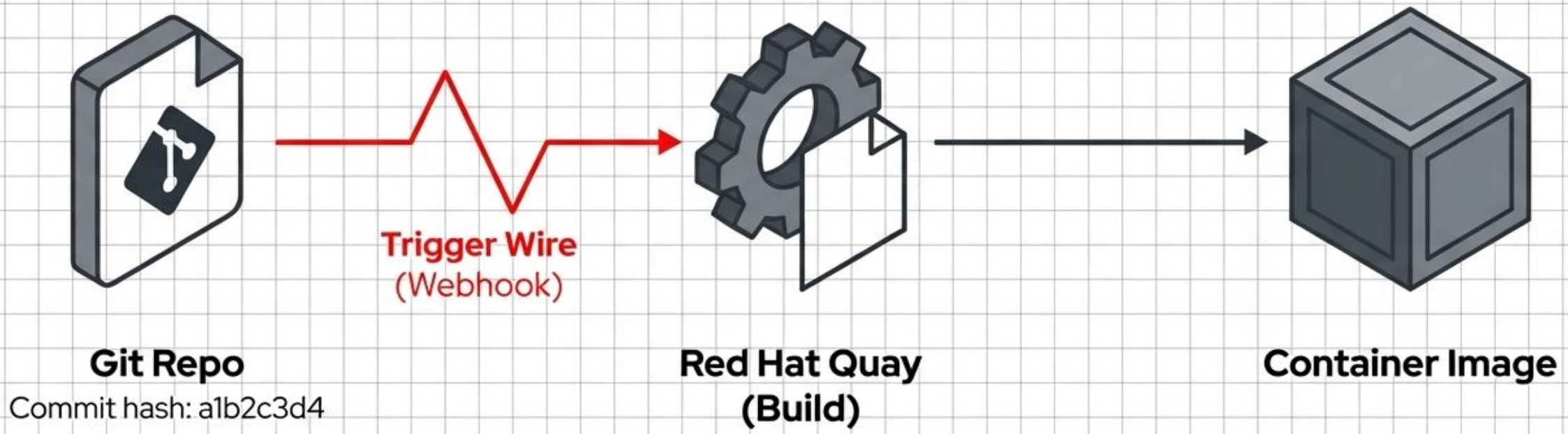
Git build triggers automate the path from code commit to container



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Summary

- Red Hat Quay is an enterprise-grade registry with RBAC, scanning, and replication
- Runs on OpenShift or other Kubernetes with supported storage backends
- Security combines identity, policy, and integration with CI/CD and admission
- Plan HA, backup, and multi-site replication for production scale
- See Red Hat Quay documentation for version-specific installation and features

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Secure the source to empower the cluster.

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Thank You

Questions? See Red Hat Quay product documentation.

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